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Contributors

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Reproducible floating-point results

1. Abstract

The C++ standard, hereafter referred to as [\[C++\]](#), has very little to say about floating-point specification, which means that it is extremely hard to write floating-point expressions that, given the same inputs and rounding mode, will produce the same results on all implementations. This paper proposes the introduction of a new type which specifies sufficient conformance with the provisions specified in ISO/IEC 60559:2020, hereafter referred to as [\[559\]](#), to guarantee this reproducibility.

2. Motivation

From [\[559\]](#) Clause 11:

Programs that can be reliably translated into an explicit or implicit sequence of reproducible operations on reproducible formats produce reproducible results. That is, the same numerical and reproducible status flag results are produced.

Reproducible results require cooperation from language standards, language processors and users. A language standard should support reproducible programming.

C++ does not support reproducible programming. In fact, the standard does not specify floating point behaviour at all. In [\[C++\]](#), [basic.fundamental] paragraph 12:

Except as specified in 6.8.3, the object and value representations and accuracy of operations of floating-point types are implementation-defined.

Even in 6.8.3 [basic.extended.fp], we read: